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Ultra Short Bowel Syndrome: Living on the Edge

Osman Mohamed Elfadil[1] · Ryan T. Hurt[1,2,3] · Manpreet S. Mundi[1] · Bradley R. Salonen[2] · Sara L. Bonnes[2] · Robert G. Martindale[4] · Jithinraj Edakkanambeth Varayil[5]

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Abstract

Purpose of Review Ultra-short bowel syndrome (USBS) is a rare and severe form of short bowel syndrome (SBS), characterized by a markedly reduced length of functional small intestine (less than 50 cm of small bowel remaining). This condition leads to profound malabsorption, fluid and electrolyte imbalances, and significant nutritional deficiencies, often necessitating long-term home parenteral nutrition (HPN).

Recent Findings USBS may arise from diverse etiologies, including congenital anomalies in children, ischemic or inflammatory diseases in adults, and USBS resulting from trauma or injury to the vascular supply during surgery. Despite advances in HPN formulations, surgical techniques, and pharmacologic agents such as teduglutide, many patients remain HPN-dependent. Multidisciplinary intestinal rehabilitation is a key strategy to promote intestinal adaptation and reduce complications.

Summary This review synthesizes current literature on epidemiology, clinical presentation, and management of USBS in pediatric and adult populations. Additionally, we present a single-center case series of adults with USBS managed through a specialized HPN program. These findings highlight the need for tailored, interdisciplinary care to optimize outcomes in this complex condition.

Keywords SBS · Short bowel syndrome · Ultra-short bowel syndrome · Intestinal failure · Parenteral nutrition · Home parenteral nutrition · HPN · PN

Introduction

Short bowel syndrome (SBS) is a complex clinical condition that leads to significant health challenges due to severe intestinal failure. SBS generally arises from considerable surgical removal of the intestine, congenital anomalies, or disease-associated loss of function, resulting in difficulties absorbing both macro- and micro-nutrients. Therefore, SBS is clinically characterized predominantly by the malabsorption of nutrients, fluids, and electrolytes, resulting in malnutrition [1]. SBS often necessitates long-term parenteral support to maintain fluid, electrolyte, and nutritional homeostasis. Although a rare disease in adults, the incidence of SBS is estimated at 0.6 per 100,000 [2, 3]. It can be more common in the pediatric population, with an incidence of SBS estimated to be 0.25 cases per 1000 live births [4–6] especially among infants with low birth weight (< 1500 g), where the incidence can be as high as 7 cases per 1000 live births [2, 3, 5]. Overall, SBS remains a leading cause of chronic intestinal failure (CIF), accounting for about two-thirds of the cases of CIF in adults [2], and more than half of the CIF cases in children [7]. The clinical presentation and long-term outcomes of SBS are heterogeneous and depend primarily on the extent and configuration of the remaining bowel.

The predominant causes of SBS vary by age group, with necrotizing enterocolitis and congenital anomalies common in children, while Crohn's disease, mesenteric ischemia, and malignancy are leading causes in adults [2–4]. With

✉ Jithinraj Edakkanambeth Varayil
EdakkanambethVarayil.Jithinraj@mayo.edu

1 Division of Endocrinology, Diabetes, Metabolism and Nutrition, Mayo Clinic, Rochester, MN, USA

2 Division of General Internal Medicine, Mayo Clinic, Rochester, MN, USA

3 Division of Gastroenterology and Hepatology, Mayo Clinic, Rochester, MN, USA

4 Division of Gastrointestinal and General Surgery, Department of Surgery, School of Medicine, Oregon Health & Science University, Portland, OR, USA

5 Department of Family Medicine, Mayo Clinic, Rochester, MN, USA

recent advancements in surgical techniques and improvements in parenteral nutrition, particularly with the introduction of newer formulations and lipids, the survival rates for patients with SBS have significantly improved [8–10]. Between 2005 and 2014, although SBS-related hospitalizations in the United States increased by 55%, the in-hospital mortality related to SBS reportedly declined significantly [11]. However, SBS presents significant challenges and morbidity due to its complex nature, often resulting in residual intestinal dysfunction and malabsorption. Some patients may attain enteral autonomy after total parenteral nutrition (PN) [12]. For many individuals with SBS, long-term home parenteral nutrition (HPN) becomes essential to support their nutritional needs. In select cases, where conventional management strategies are insufficient, gut transplantation may be considered a viable option [13]. The use of long-term home enteral nutrition (HEN) in patients with SBS is very limited, with fewer than 0.5% of those receiving long-term HEN being diagnosed with SBS [14]. It is imperative that all patients diagnosed with SBS and receiving HPN continue to receive low-dose enteral nutrition whenever feasible. This approach aims to utilize the remaining gut segment, thereby preserving mucosal integrity [15, 16].

Ultra-short bowel syndrome (USBS) is a particularly severe form of SBS characterized by a highly reduced length of the functional small intestine. This dramatic loss of absorptive surface area leads to significant nutrient and fluid absorption challenges, often resulting in malnutrition, dehydration, and electrolyte imbalances. This review provides a comprehensive and up-to-date overview of the literature on this subtype, which is usually regarded as a heterogeneous group of complex and challenging clinical cases. These cases require specialized interdisciplinary care and frequently necessitate prolonged HPN or gut transplantation. The incidence and prevalence of USBS in adults and children are not well-established in literature. In addition to the comprehensive literature and evidence review, the authors present a single-center experience through a case series of PN data of adult patients diagnosed with USBS. These patients were followed up in a specialized interdisciplinary HPN program at a quaternary care academic center.

Definitions and Classifications

The definition of SBS is based on the length of the remaining small intestine measured from the ligament of Treitz at the duodenojejunal flexure. A shorter length of the small intestine is associated with poorer outcomes and higher mortality, primarily due to complications related to CIF and PN [11]. Other factors contributing to the prognosis of patients with SBS include the underlying disease, other metabolic co-morbidities and the degree of compensatory reaction of the remnant small bowel [3].

SBS is typically defined by having a functional small intestine measuring less than 200 cm from the ligament of Treitz in adults [3, 17]. In children, SBS is defined as a remaining small intestine length that is less than 25% of the length predicted for their age or when the need for PN persists for more than 2 months following intestinal surgery [4, 6]. This definition has now been added to the International Classification of Diseases, 11th Revision. The term SBS was first introduced in 1962 to highlight the observation that the outcomes of intestinal resection depended on the length of the remaining small intestine [3]. Before this, studies discussing the effects of extensive bowel resections focused on the length of bowel removed rather than the portion that remained [3]. Since then, SBS has been classified through several frameworks, each emphasizing different dimensions of the syndrome's clinical and pathophysiological complexity. It is now crucial that surgical and medical records aggressively document the remaining length of small bowel and not just what was removed at surgery.

Anatomically, SBS is categorized by the length and specific segments of the remnant bowel, particularly the presence or absence of the ileum and colon, including the presence or absence of the ileocecal valve. This classification stratifies patients into two principal categories based on colonic continuity: those with a jejunostomy and no colon in continuity (type 1) and those with an intact colon in continuity (types 2 and 3). Following surgical resection, these anatomical subtypes exhibit distinct differences in three critical aspects of intestinal function: the production of short-chain fatty acids (SCFAs) by the colonic microbiota, the secretion of gastrointestinal hormones, and the capacity for water and sodium absorption [3]. This is also very significant from a pathophysiological standpoint. Patients with colon continuity often exhibit improved hydration status and increased potential for weaning from intravenous support (IVS) due partly to colonic bacterial fermentation of carbohydrates not digested and absorbed in the small intestine into short-chain fatty acids, which can contribute to a salvage of 700–1000 kcal/day [18]. In contrast, those with end-jejunostomies are at greater risk of high-output states and prolonged dependence on PN support [11]. Clinically, SBS may be stratified by functional capacity into intestinal insufficiency, where partial enteral absorption remains feasible without IVS, and intestinal failure, where IVS is indispensable for survival.

Ultrashort bowel syndrome (USBS) is a specific subtype of SBS characterized in adults by a remnant small bowel length of less than 50 cm [19], and in children by a remnant length that is less than 10% of the expected length for their age [5, 6, 20]. The incidence and prevalence of USBS in adults and children are challenging due to the condition's rarity and the variability in the definitions used in the literature.

USBS in Adults

Patients with USBS frequently present with symptoms related to malabsorption, including diarrhea and steatorrhea, dehydration and electrolyte imbalances, weight loss, and malnutrition [21]. In addition to that, these patients often also experience vitamin and mineral deficiencies such as in vitamin B12, fat-soluble vitamins, calcium, magnesium, and zinc. Furthermore, metabolic acidosis, gastric acid hypersecretion, and complications such as renal oxalate and cholesterol biliary calculi are also common in patients with USBS [19, 22]. The severity of these symptoms is often proportional to the extent of bowel loss and the specific functional capacity of the remaining intestinal segment [20, 21]. Additionally, psychological impacts, such as anxiety and depression, can arise from the chronic nature of the condition and the burdens associated with long-term management [23]. The diagnosis of USBS is primarily based on clinical history, imaging studies, and dietary assessments, as accurate bowel length is often not measured or recorded in operative notes after surgical resection. Using the pathology reports based on formalin fixed tissue is not accurate as tissue shrinks in size dramatically during storage in chemicals used for preservation. Precise measurement of the remaining bowel length and documenting at the time of resection immediately after surgery is crucial, as insufficient documentation can result in a lack of coverage of lifesaving therapies such as PN. Without measurements in the OR at the time of resection, estimating bowel length can be attempted using imaging modalities such as MRI or CT combined with nutritional assessments and laboratory tests (such as citrulline) to define USBS [19, 24]. Further evaluations may include nutrition assessments and laboratory tests to identify any deficiencies or complications arising from malabsorption.

USBS management necessitates a multidisciplinary approach tailored to individual patient needs. Nutritional support is foundational, with PN often required initially to meet caloric, fluid, and electrolyte demands. Transition to EN and oral intake is pursued as feasible to encourage intestinal adaptation. However, most patients with USBS remain PN-dependent for life. Pharmacologic therapy includes antidiarrheal agents (e.g., an opium tincture, loperamide, or codeine phosphate) to reduce the transit time, antisecretory drugs (e.g., H2 antagonists, proton pump inhibitors) to control gastric hypersecretion, and teduglutide, a glucagon-like peptide-2 analog that stimulates hypertrophy of mucosa and enhances nutrient, electrolyte and fluid absorption and decreases reliance on parenteral support. Fluid and electrolyte balance is maintained through oral rehydration solutions or intravenous fluids, particularly in patients with high-output stomas or severe dehydration. In selected cases where conventional therapies are insufficient, surgical options to lengthen the remaining small intestine may be considered but rarely used in adults. Intestinal transplantation and reconstruction, when other options have failed [25–27]. This approach can restore gastrointestinal function in patients with severe nutrient malabsorption and intractable complications. It is also vital to address the psychological and social aspects of living with USBS in a timely and effective manner.

USBS in Children

Similar to adults, USBS children commonly present with chronic diarrhea, failure to thrive, dehydration, electrolyte imbalances, and nutrient deficiencies [20, 28]. Clinical features often include poor weight gain, growth failure, and signs of intestinal dysmotility such as abdominal distension and vomiting. Long-term PN is the cornerstone of treatment, providing essential nutrients intravenously to support growth and development during intestinal adaptation [29]. Early and progressive initiation of enteral nutrition is crucial to stimulate adaptation, with breast milk being the preferred source; however, extensively hydrolyzed or amino acid-based formulas may be used when necessary [30, 31]. Oral feeding should also be encouraged to prevent oral aversion [32]. Pharmacologic interventions manage symptoms and support adaptation, including antidiarrheal agents, acid suppressants, bile acid binding resins, and enteral antibiotics. Teduglutide, a glucagon-like peptide-2 (GLP-2) analog, has shown promise in enhancing mucosal growth and enteral tolerance [33–36]. GLP-2 plays a role in the growth of the intestine and can promote both hypertrophy and hyperplasia in the small intestine. This effect may be advantageous for individuals with USBS, where the absorptive surface of the small intestine is diminished. GLP-2 encourages the proliferation of ileal cells and decreases apoptosis, resulting in a larger intestinal mass and greater absorptive surface area [37, 38]. Surgical interventions, such as bowel lengthening procedures (e.g., serial transverse enteroplasty) and restoration of intestinal continuity, may further improve nutrient absorption and reduce dependence on PN [12]. In cases where complications from PN become irreversible, intestinal transplantation may be considered. Optimal outcomes are achieved through care delivered by a specialized multidisciplinary intestinal rehabilitation team comprising pediatric gastroenterologists, surgeons, dietitians, and other healthcare professionals. The overarching goal of management is to promote enteral autonomy while minimizing complications associated with long-term PN, such as intestinal failure-associated liver disease and catheter-related bloodstream infections.

Observations from Clinical Experience

In this section, we present our clinical observations derived from a cohort of seven patients diagnosed with USBS, drawn from the prospectively maintained Mayo Clinic HPN program dataset. This group was matched with a control cohort of equivalent size, ensuring an identical gender composition and comparable age and anthropometrics at the initiation of HPN. The control group includes patients with SBS with a presumed remnant gut length of at least 100 cm. It is imperative to note that these findings should not be construed as formal published data or as a basis for comparative conclusions; instead, they serve as descriptive clinical insights into the clinical outcomes associated with HPN in patients with USBS. Notably, the remnant small bowel length averages show a stark difference; patients with USBS had a shorter remnant (28 cm) than those with SBS (111 cm).

Table 1 outlines the demographic and clinical characteristics of patients with SBS receiving HPN. Notably, patients with USBS tolerated and benefited from HPN for a mean of 11 years, indicating that patients with no ultrashort gut can be supported with parenteral nutrition therapy for an extended period, similar to their counterparts with longer remnant gut. Interestingly, the table details the primary diagnoses leading to SBS, with inflammatory bowel disease (IBD) being the most common, followed by bowel ischemia and malignancy. This information highlights the diverse etiologies of SBS, reinforcing the need for tailored treatment plans based on individual patient backgrounds.

Table 2 presents HPN utilization in patients with USBS, providing insight into the frequency and duration of therapy. On average, USBS patients receive HPN daily, similar to the general SBS. It is important to note that patients with USBS needed higher base and total HPN volumes as expected. However, they have achieved fewer calories of the calculated needs based on the Harris-Benedict formula, with patients with USBS achieving a median of 78% of the target compared to 100% in the matched SBS cohort.

Table 1 Demographic and clinical characteristics of patients with SBS receiving HPN

Characteristics Mean (SD)	USBS n = 7	SBS n = 7
Demographics		
Age at HPN initiation, y	53 (21)	53 (20)
Current age/ age at end of HPN, y	63 (18)	61 (19)
Female: Male	4:3	4:3
Anthropometrics		
Weight at PN initiation, kg	63 (14)	63 (21)
BMI at HPN initiation, kg/m^2	21.8 (2.7)	21.7 (3.7)
Remnant small bowel length, cm	28 (14)	111 (19)
Primary diagnosis, n		
IBD	3	3
Bowel ischemia	1	1
Malignancy	1	1
Malabsorption	1	1
Bariatric surgery	1	–
Radiation enteritis	–	1
PN access	Single Lumen PICC (1)	Single Lumen PICC (1)
	Single Lumen Hickman (3)	Single Lumen Hickman (1)
	IVAD (Port-a-Cath/SL PORT/DL Port) (2)	Unknown (5)
	Double Lumen Hickman (1)	
Colon continuity, n		
Yes	1	3
No	6	4
Duration on HPN, y	11 (4)	8 (5)

Table 2 HPN Utilization in Patients with SBS

Characteristics Mean (SD) Median (IQR 25th, 75th)	USBS n = 7	SBS n = 7
HPN utilization		
HPN days per week, d	6 (1)	7 (0)
Infusion duration, h	12 (2)	12 (2)
Achieving nutrition goals		
Harris-Benedict (HB) calories, kcal/d	1306 (241)	1486 (427)
% HB PN delivered, %	78 (72,124)	100 (64,128)
HPN formula		
Total volume, L	3 (2,3.8)	2.25 (1.25,3)
Base volume, L	2.75 (2,3.6)	2 (1,2.5)
Daily average PN calories, kcal	1088 (314)	1163 (427)
Weekly average PN calories, kcal	1325 (490)	1278 (405)
Amino acids, g	95 (36)	68 (18)
Lipid emulsion		
Use of lipids, n	6	7
ILE days per week, d	3.8 (2)	2 (2)
Lipids, g	48 (8)	52 (23)
Lipid calories, kcal	433 (82)	510 (140

Conclusion

In conclusion, USBS represents one of the most severe phenotypes within the spectrum of SBS, characterized by profound reductions in intestinal length and associated with significant clinical, nutritional, and psychosocial challenges. Both pediatric and adult patients with USBS require highly individualized, multidisciplinary care plans that address the complex interplay of malabsorption, fluid and electrolyte imbalances, nutritional deficiencies, and intestinal adaptation. Despite recent advances in medical, surgical, and pharmacological therapies that include teduglutide and bowel-lengthening procedures, many patients continue to depend on long-term HPN, which carries its risks and burdens. When feasible, early and progressive enteral feeding remains critical to stimulate adaptation and support weaning from parenteral support. This review highlights the importance of specialized intestinal rehabilitation programs in optimizing outcomes and minimizing complications in this high-risk population. By integrating evidence from the current literature with clinical insights from single-center experience, this work contributes to a more nuanced understanding of USBS. It underscores the need for continued innovation and research to improve long-term outcomes and quality of life for affected individuals.

Key References

Papers of particular interest, published within the annual period of review, have been highlighted as: (*) of special interest (**) of outstanding interest.

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* This review provides essential information on the latest classification of SBS and its clinical significance.

2. ** L. Norsa et al., “Nutrition and Intestinal Rehabilitation of Children With Short Bowel Syndrome: A Position Paper of the ESPGHAN Committee on Nutrition. Part 1: From Intestinal Resection to Home Discharge,” J. Pediatr. Gastroenterol. Nutr., vol. 77, no. 2, pp. 281–297, Aug. 2023, doi: https://doi.org/10.1097/MPG.0000000000003849.

** This recent position paper provides essential insights on intestinal rehabilitation and nutrition support for children with SBS.

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Declarations

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